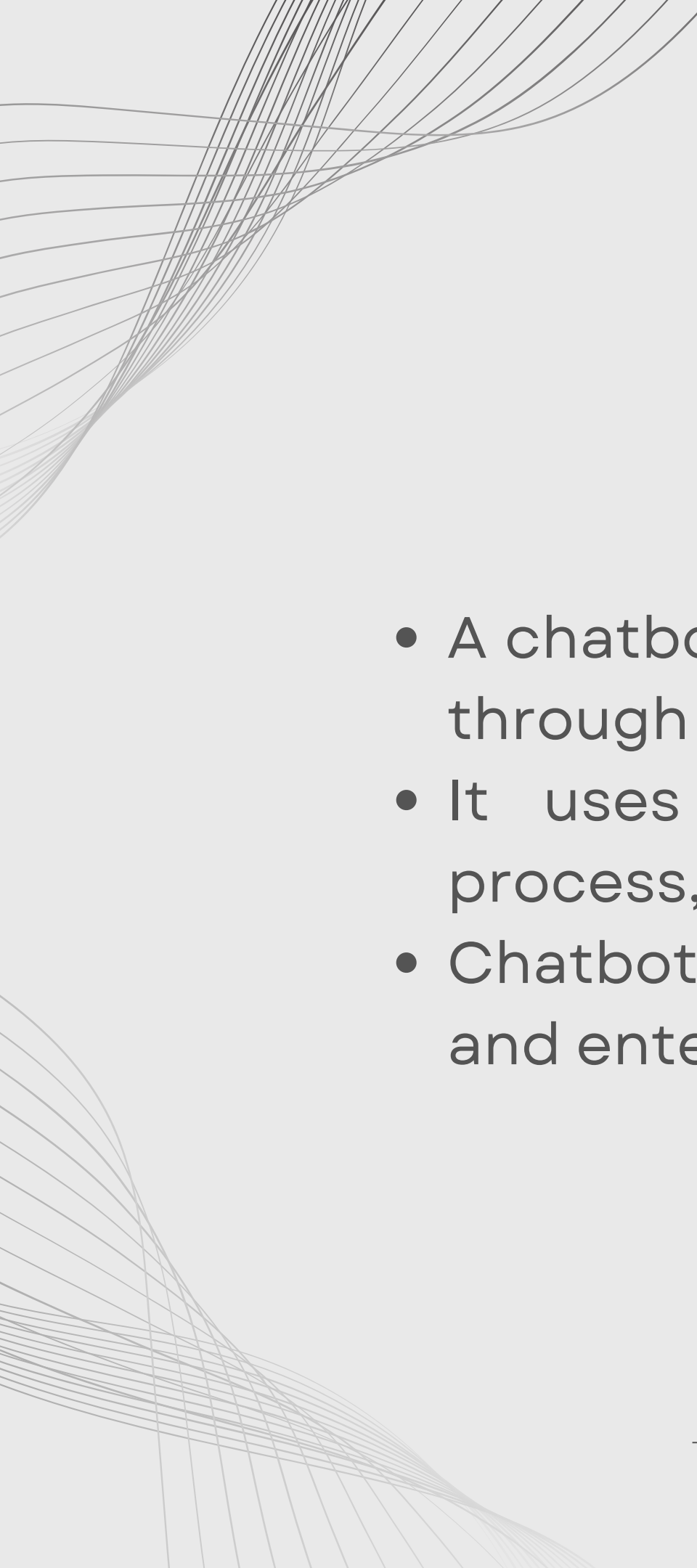




CHATBOTS

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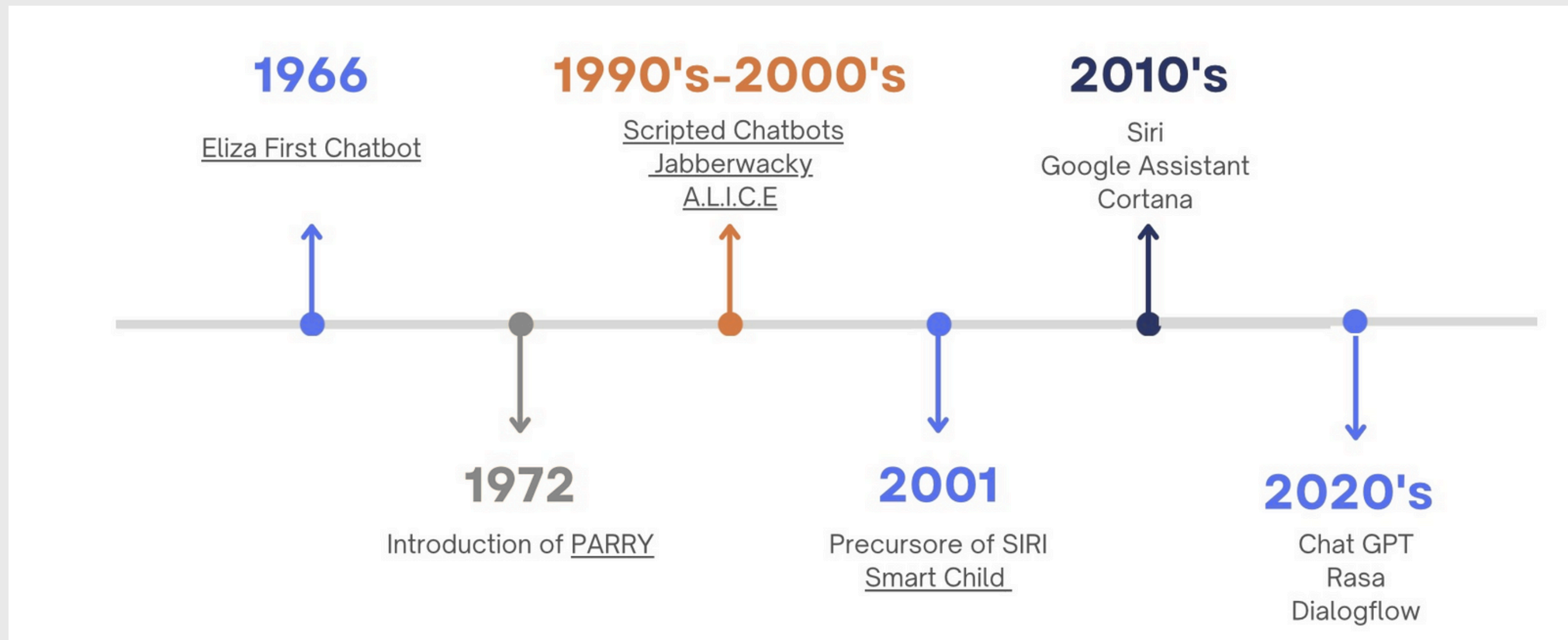


INTRODUCTION

What are Chatbots?

- A chatbot is an AI program that can simulate human conversation through text or voice.
- It uses Natural Language Processing (NLP) to understand, process, and respond intelligently.
- Chatbots are used in customer service, education, healthcare, and entertainment.

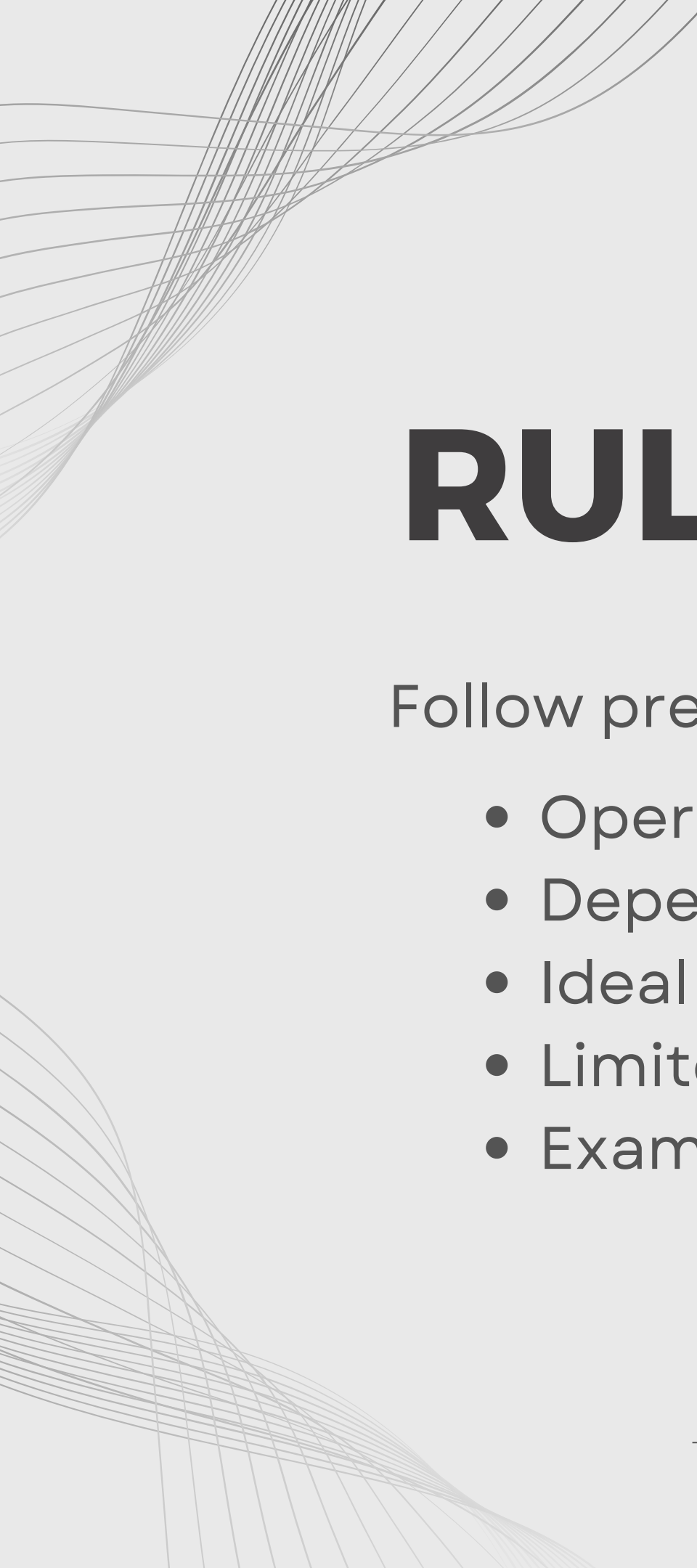
HISTORY OF CHATBOT



TYPES OF CHATBOTS

Chatbots can generally be divided into four main types, based on how they process and respond to user input.

- **Rule-Based Chatbots** – Follow predefined scripts and keywords.
- **Retrieval-Based Chatbots** – Use machine learning and NLP to understand intent.
- **AI-Powered Chatbots** – Combine NLP, ML, and contextual understanding to deliver personalized, adaptive responses.
- **Generative Chatbots** – Use deep learning and large language models for natural, context-aware conversations.
- **Voice-Based Chatbots** – A chatbot that accepts speech, processes it, and responds using generated speech.



RULE-BASED CHATBOTS

Follow predefined rules or decision trees to respond to user inputs.

- Operate on if-then rules or decision trees.
- Depend on keywords or patterns.
- Ideal for simple and repetitive tasks such as FAQs.
- Limited in understanding context or unexpected inputs.
- Example: FAQ bots, Ticket booking etc.

Examples

Follow predefined “If user says X → reply Y” rules.

Basic Example	User: "Show me the restaurant menu." Bot: "Sure! Type 1 for Starters, 2 for Main Course, 3 for Desserts." If you say something unexpected like "I'm hungry", it may reply: "Invalid input. Choose 1, 2 or 3."
Technical Example	Uses simple if-else keyword rules to return predefined answers. If the user input is not a choice, prompt the user with the initial choices again.

RETRIEVAL-BASED CHATBOTS

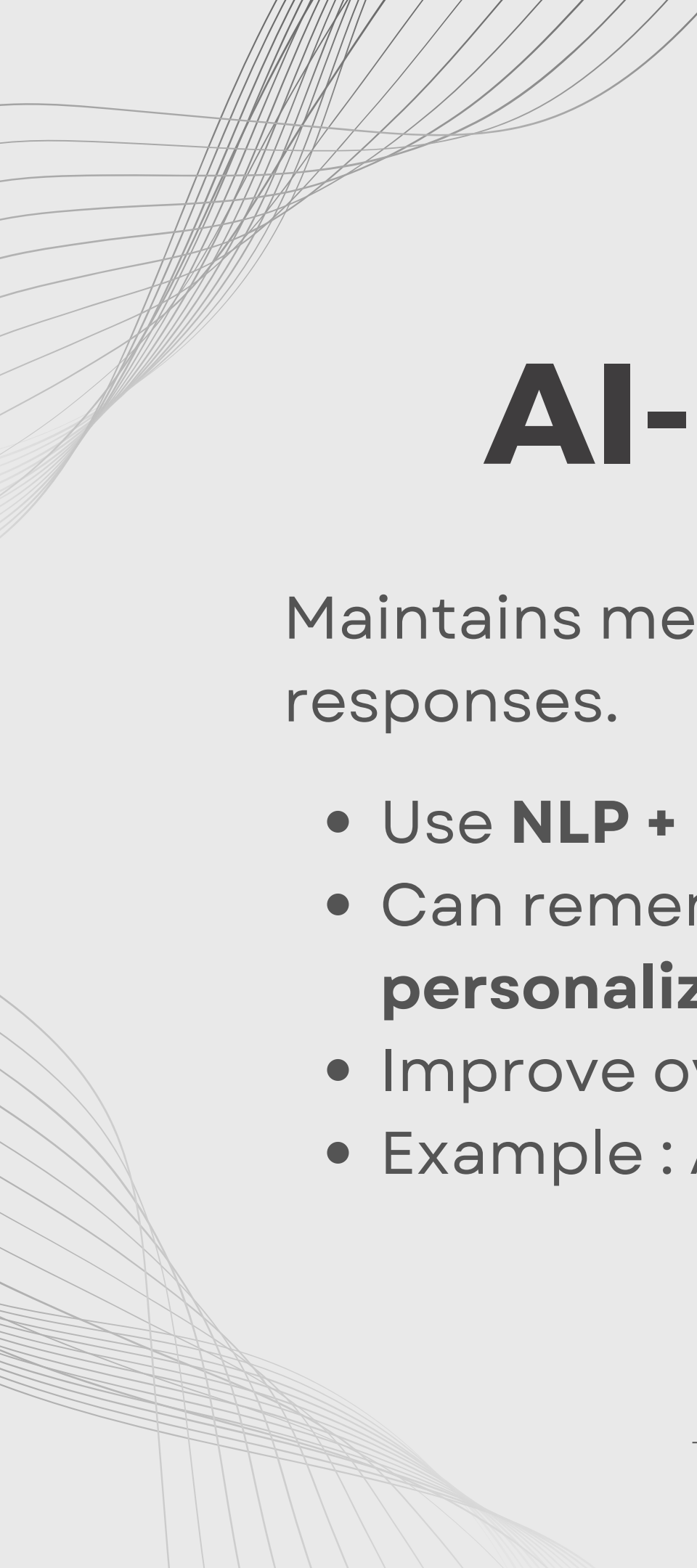
Use NLP to find the best-matching predefined response from a knowledge base.

- Understand what the user means using NLP. Can analyze inputs rather than just looking for keywords.
- Pick the best-matching answer from a list of stored responses.
- Can maintain **short-term context** in a conversation.
- Cannot create new answers – only choose from what's already saved.
- Example : Basic customer support bots.

Examples

Uses NLP to understand queries but still picks responses from a stored database.

Basic Example	User: "I lost my password." Bot: "You can reset it using the Reset Password link." Even if user types: "Forgot password", "Can't login", "Need password help", bot finds the closest matching stored response.
Technical Example	Uses intent detection, embeddings, or similarity matching.



AI-POWERED CHATBOTS

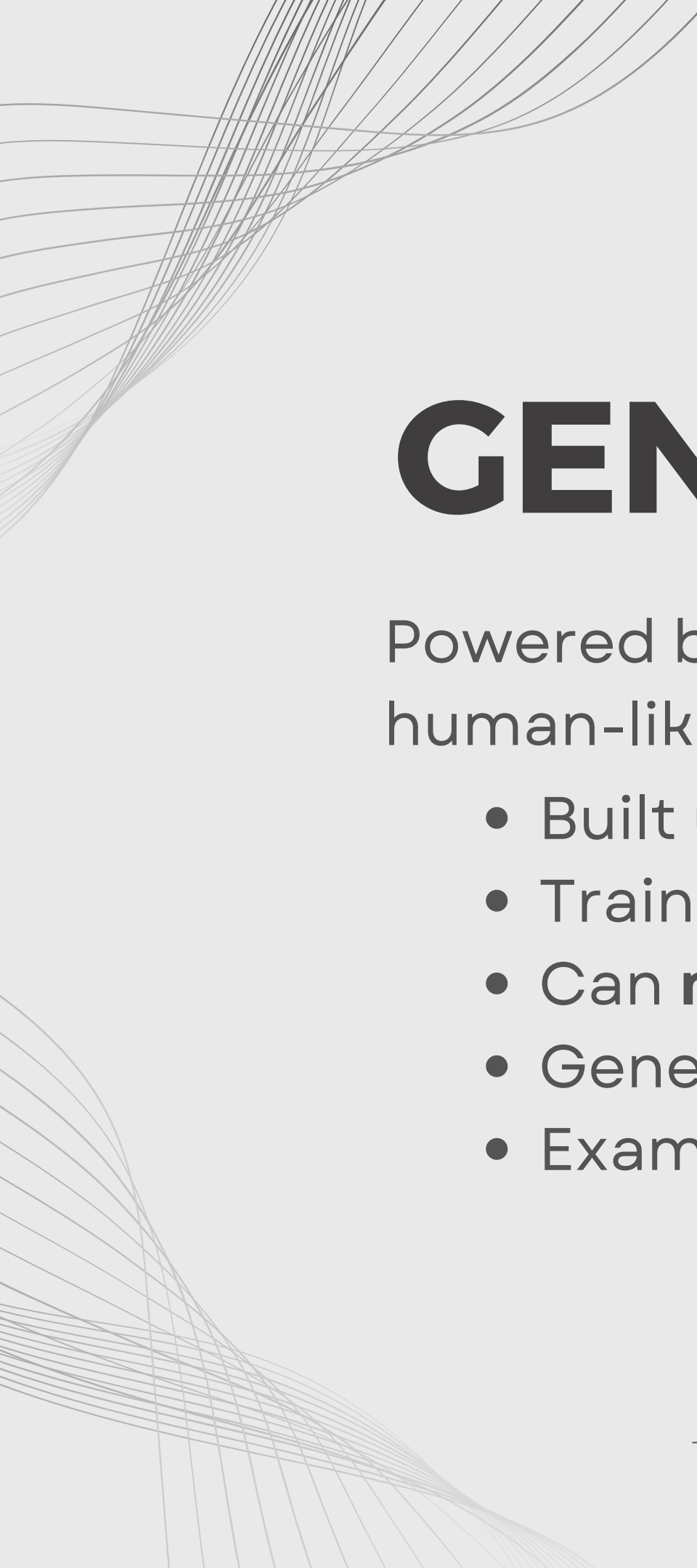
Maintains memory of past interactions to give personalized, context-aware responses.

- Use **NLP + machine learning** to understand intent and context.
- Can remember user preferences or conversation history to give **personalized** responses.
- Improve over time through continuous learning.
- Example : Alexa, Siri, Google Assistant.

Examples

Learns from conversations, remembers preferences, understands intent, and improves over time.

Basic Example	User: "Book a cab to my office." Bot: "Sure! Same office as yesterday? Kochi Infopark?" Bot remembers past interactions and personal preferences.
Technical Example	Uses NLU engines, intent classification, entity extraction, and sometimes user profiling to personalize responses.



GENERATIVE CHATBOTS

Powered by Large Language Models (LLMs) that generate original, human-like text.

- Built using **transformer architecture** and deep learning.
- Trained on vast datasets for grammar, facts, and reasoning.
- Can **manage complex, multi-turn conversations** seamlessly.
- Generates **creative, contextually relevant** responses.
- Examples: ChatGPT, Gemini, Claude.

Examples

Doesn't rely on predefined responses – generates new sentences like a human.

Basic Example	User: "Tell me a bedtime story about a robot and a cat." Bot: (creates a completely new story never stored anywhere.)
Technical Example	Uses Transformer models (GPT, Gemini, Claude) trained on massive datasets. Example inference: text → tokenization → transformer layers → next token prediction → output response

ADDITIONAL: VOICE-BASED CHATBOTS (VOICE ASSISTANTS)

A chatbot that listens to speech input, converts it into text, processes the meaning, and responds using synthesized speech output.

- Uses Automatic Speech Recognition (ASR) to convert spoken audio into text.
- Processes the text using NLP and intent detection to understand the command.
- Uses a Text-to-Speech (TTS) engine to speak the response back to the user.
- Can perform hands-free tasks such as answering questions, controlling devices, or assisting with navigation.
- Commonly used in smartphones, smart speakers, and IoT devices.
- Examples: Google Assistant, Amazon Alexa, Apple Siri.

Examples

A chatbot that accepts speech, processes it, and responds using generated speech.

Basic Example	User: "Hey Google, turn on the lights." Assistant: "Okay, turning on lights."
Technical Example	• Speech-to-Text: When the user says: "Turn off the lights", the system records the audio and converts it into text. • Intent Recognition: The system identifies that the user wants to perform a home automation command (turning lights off). • Action + Response: The assistant executes the command and prepares a response such as: "Okay, turning off the lights." • Text-to-Speech: The response is spoken out loud to the user.

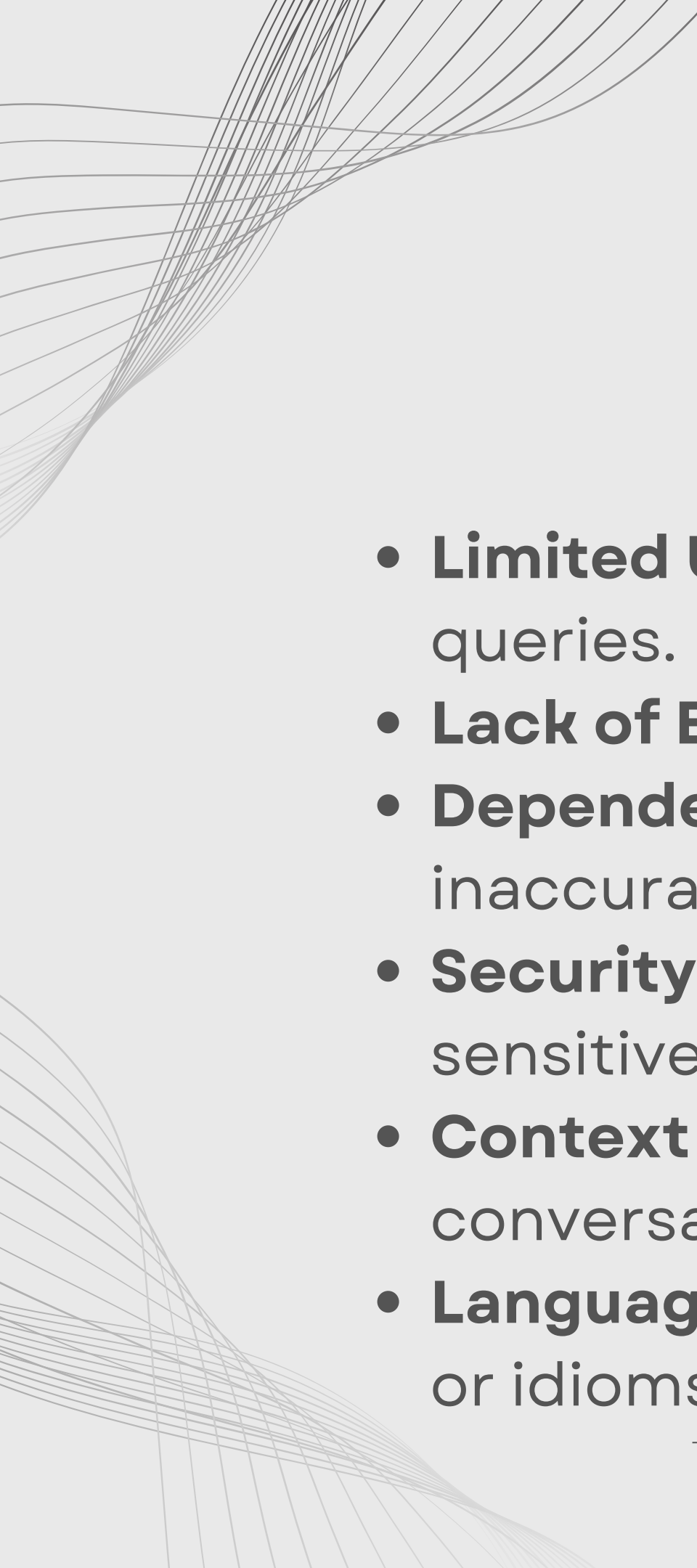
Summary Table

<u>Type</u>	<u>Understanding</u>	<u>Response Type</u>	<u>Strength</u>	<u>Weakness</u>
Rule-Based	Keyword matching	Fixed	Fast, predictable	No learning
Retrieval-Based	NLP + similarity matching	Best matching stored text	Handles variations well	Cannot create new replies
AI-Powered	ML + memory	Adaptive + personalized	Learns user behavior	Needs training + data
Generative	Deep Learning (LLM)	Fully generated	Human-like	Requires high compute
Voice-Based	ASR + NLP + TTS	Spoken	Hands-free interaction	Requires speech models



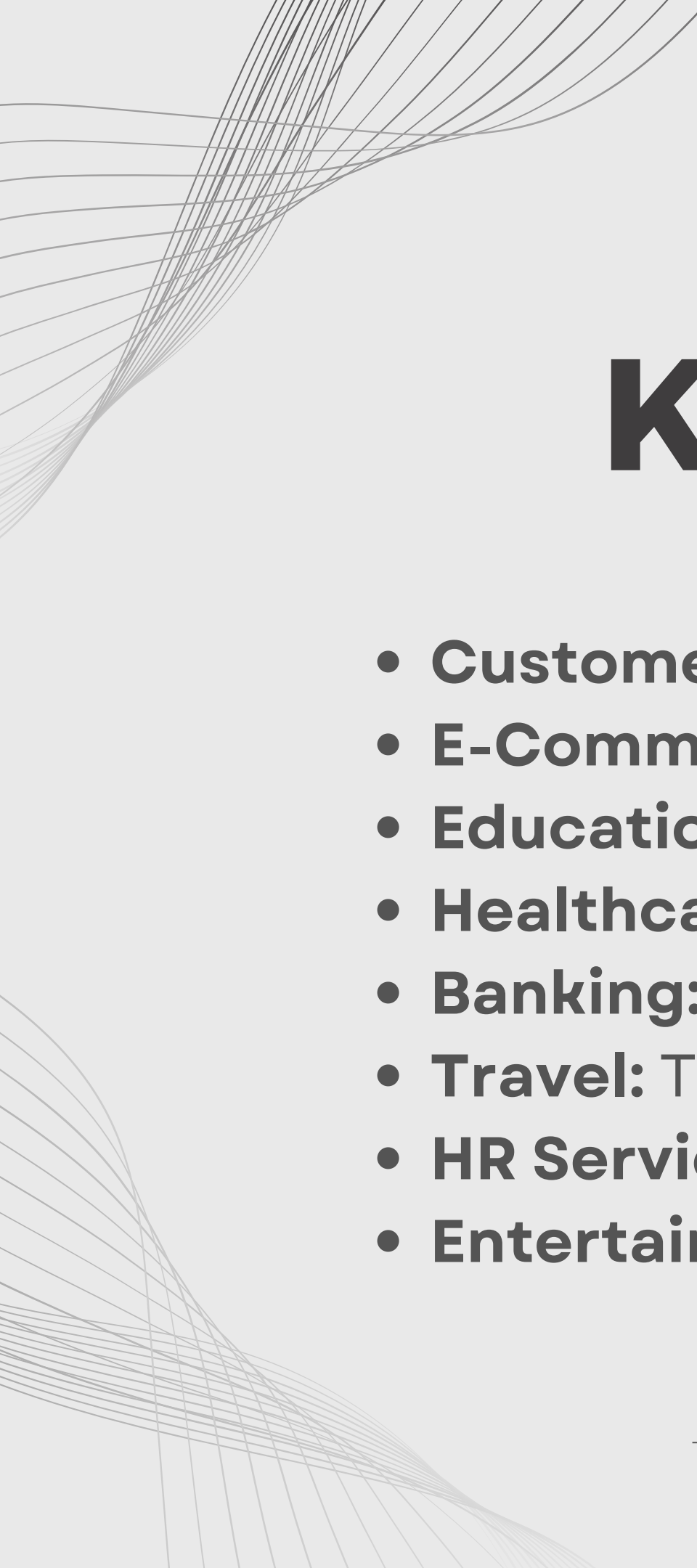
ADVANTAGES

- **24/7 Availability:** Provide continuous support without human intervention.
- **Cost-Effective:** Reduce the need for large customer support teams.
- **Instant Responses:** Handle multiple queries simultaneously.
- **Personalization:** Offer customized experiences based on user data.
- **Scalability:** Can easily handle growing user demand.
- **Data Collection:** Gather insights for business analytics and improvement.



CHALLENGES

- **Limited Understanding:** Struggle with ambiguous or complex queries.
- **Lack of Emotion:** Cannot fully replicate human empathy or tone.
- **Dependence on Training Data:** Poor data quality can lead to inaccurate responses.
- **Security & Privacy Concerns:** Risk of data breaches or misuse of sensitive information.
- **Context Retention:** Difficult to maintain long-term conversational memory.
- **Language & Cultural Barriers:** May misinterpret regional phrases or idioms



KEY APPLICATIONS

- **Customer Support:** 24/7 automated assistance for user queries.
- **E-Commerce:** Product recommendations and order tracking.
- **Education:** Virtual tutors for learning support.
- **Healthcare:** Appointment booking and basic medical guidance.
- **Banking:** Balance checks, transaction info, and alerts.
- **Travel:** Ticket booking and trip assistance.
- **HR Services:** Employee query handling and onboarding.
- **Entertainment:** Personalized interaction and content delivery.



CONCLUSION

- Chatbots have revolutionized the way humans interact with digital systems.
- Powered by AI, NLP, and Large Language Models (LLMs), they are evolving from simple rule-based programs to intelligent, context-aware assistants.
- They enhance efficiency, reduce costs, and improve user experience across various sectors.
- As technology advances, chatbots will continue to become more natural, adaptive, and human-like, shaping the future of human-computer interaction.

The image features abstract line art in the top-left and bottom-left corners. These elements consist of numerous thin, dark grey lines that curve and sweep across the page, creating a sense of movement and depth. The lines are more densely packed in some areas, forming soft, cloud-like shapes, while in others, they are more sparse and delicate.

THANK YOU
